

Evaluation of LABOR Synthesizer for Educational Purposes in Electronics

Punnavarjula Sai Harshita

October 2026



Faculty of Electrical Engineering and Computer Science

Evaluation of LABOR Synthesizer for Educational Purposes in Electronics

by

Punnavarjula Sai Harshita

Matr.-No.:

A Thesis presented to the City University Bremen
in fulfillment of the thesis requirement for the degree of
Master in Electronics Engineering

1. Examiner: Prof. Dr.-Ing. M. Meiners
2. Examiner: Prof. Dr.-Ing. M. Goldenbaum

Preparation Period: 10.2026 – 05.2027

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Preface

This thesis template is intended for honours, masters or PhD students at the Australian National University (ANU) who wish to write their thesis using the [Quarto](#) document format. It is highly recommended for students who code using Python, R or Julia and have many computational or analysis results in their thesis.

i Note

This thesis template is available on GitHub at github.com/anuopensci/quarto-anu-thesis.

Benefits

The benefits of using Quarto document include:

- It allows you to write your thesis in a simple markup language called [Markdown](#). This means that you can focus on writing your thesis without having to worry about formatting.
- The document can be output to a variety of formats including PDF, HTML, Word, and LaTeX.
- Code can be easily embedded in the document and executed. This means that you can include the results of your analysis in your thesis without having to manually copy and paste them. This is a good reproducible and scientific practice.
- You can easily integrate with aspects of GitHub (edit, reporting an issue, etc).

The above outlined benefits can also be considered as best practice. Version controlling and collaborative writing (via Git and GitHub) are important in managing multiple versions of your thesis and in collaborating with your supervisory team. Embedding code in your thesis is a good practice in reproducible research. Making your thesis in HTML format can allow for interactive web elements to be embedded while PDF format can be for general distribution and printing.

Getting started

There are several systems that you are expected to know to use this template. These include:

- Markdown syntax for writing
- Quarto or R Markdown syntax (note that these works for Python or Julia too) for embedding code
- (Optional) Git and GitHub for hosting

Frequently asked questions

What about Overleaf?

ANU has a professional account for Overleaf, which is great for those that use LaTeX regularly. Unfortunately, there is no equivalent system with track changes in Quarto. You can output the tex file from Quarto document and use this in Overleaf. The changes made in this tex document however has to be manually transferred back to the Quarto document. If your main output is mainly mathematical and you have little to no code outputs, Overleaf is probably a better choice.

Declaration of Authorship

https://www.hs-bremen.de/assets/hsb/de/Dokumente/ZLL/MMCC/declaration_indep_work_No_v_2024_Form.pdf

I hereby certify that I have written this thesis independently and have not used any sources or aids other than those specified. Sources and aids other than those indicated. The passages in the thesis that have been taken from other works in terms of wording or meaning are from other works are identified as such.

This statement also applies to the graphics, sketches and illustrations contained in the thesis. I have not already submitted the work in the same or a similar form, including extracts, as part of an examination or coursework.

I hereby certify that the electronic version of the dissertation submitted is a complete copy of the printed version. and agree to an electronic check of the thesis using plagiarism software.

Please tick the box: One of the two options must be chosen in consultation between the assessor and the assessee.

- ☐ Option 1: Use of AI-based tools without mandatory labelling I confirm that we have not used any AI-based tools whose use has not been agreed in writing with the verifier. I understand that the use of text or other content and products generated by AI-based tools is not a guarantee of their quality. I accept responsibility for the work submitted and the content contained therein. I also assure you that my creative influence predominates in this work.
- ☐ Option 2: Use of AI-based tools with mandatory labelling I confirm that we have not used any AI-based tools whose use has not been agreed in writing with the verifier. I understand that the use of text or other content and products generated by AI-based tools is not a guarantee of their quality. I accept responsibility for the work submitted and the content contained therein. I also assure you that my creative influence predominates in this work. All verbatim or analogous adaptations and quotations, and all sections designed, written and/or edited using AI-based tools, will be marked and checked by me. The form of marking will be agreed between the examiner and the examinee.

We are aware that failure to comply with the above may have consequences in terms of examination law, and in particular may result in the examination being marked “insufficient” or the coursework being marked “failed” and, in the case of multiple or serious attempts to cheat, may result in my being de-registered.

First name and surname:

Matriculation number:

Date:

Signature:

Abstract

Acknowledgements

I would like to express my sincere gratitude to my dog, Chuckles, for eating my research notes multiple times. If it wasn't for you, I would have finished this thesis earlier.

Chapter 1.

Introduction

Synthesizer is an electronic musical instrument that generates audio signals which can be converted to sound. Synthesizers create a wide variety of sounds by manipulating electrical signals through various synthesis techniques. Analog synthesizers, in particular, use continuous voltage signals to produce sound. The design and implementation of analog synthesizer circuits require a deep understanding of electronic components, signal processing, and circuit design principles.

Instead of adding several components to a breadboard and connecting them with wires, the LABOR workstation provides a more structured and efficient environment for circuit prototyping. It allows for the integration of multiple circuit modules which includes a full-size breadboard, a dual power supply with short-circuit protection, an oscilloscope, and built-in signal and control sources and an interfacing section, enabling the creation of complex synthesizer architectures without the need for excessive wiring. This makes it an ideal platform for educational purposes, allowing students to focus on understanding the underlying principles of analog synthesis and known as “All in one workstation”.

This thesis presents the design and implementation of electronic circuit architectures for analog music synthesis using the LABOR workstation as a practical development and prototyping platform. The work integrates hardware circuit design with software components to create a functional electronic music synthesis system suitable for educational purposes particularly for observing, measuring and evaluating the generated signals [1].

The work will be inspired by Moritz Klein’s synthesizer-design tutorials and approach starts with fundamental circuit concepts and progressively develops them into functional synthesizer building blocks [2].

The goal is not only to reproduce existing synthesizer circuits but rather their underlying design ideas will be used as a starting point for developing, testing and documenting an independently designed set of circuits on the LABOR workstation.

1.1. Overview

This thesis will explore three circuit areas that can be combined to create a small experimental electronic music system. The first part will focus on converting a sawtooth waveform into a triangular waveform, with particular attention to how the circuit design and component values influence the output. The second part will involve developing a simple hardware-based sequencer for generating rhythmic patterns using clock signals, shift registers, and sequence controls. The final part will investigate the generation of different types of noise, particularly white, pink, and blue noise, and how their frequency characteristics can be shaped through filtering. After developing and testing the three circuit sections individually. @Meyer_1976 [3] [4] [5] [6]

The main purpose of the thesis is not only to build the circuits, but also gain a practical understanding of how they work and how different components and design choices affect their behaviour. The circuits will be tested in the laboratory, and the measured results will be compared with the expected theoretical behaviour. Software tools will also be used when they are useful for visualising or analysing the signals.

This approach will be expected to help students connect theoretical principles with observable sound phenomena and measurable electrical signals. In each exercise, students predict the result,

measure it, and compare it with theory. The built-in tools of the LABOR workstation supports the experimental process by minimizing potential wiring errors and enabling students to carry out the exercises with little supervision. Designing and building a circuits that generate audible signals also expected to keep students interested, engaging and motivating practical learning experience. Together, the three circuit areas teach useful concepts in operational amplifier circuits, digital logic, and filter design.

1.2. Objective

The primary objective is to develop synthesizer circuit architectures that can be implemented on the LABOR platform, encompassing core building blocks such as voltage-controlled oscillators (VCOs), envelope generators, filters, and amplifiers.

The list of tasks to be accomplished in this thesis includes:

- To understand the electronic principles behind waveform conversion, sequencing and noise generation
- To design and implement the circuits on the LABOR workstation
- To test and evaluate the circuits, comparing measured results with theoretical expectations
- To develop hardware /software interfaces for testing and analysing the circuits
- To document the design process, circuit schematics, and experimental results in a clear and educational manner

1.3. Work Packages

1. Technical research and literature review:

This phase will focus on building a solid understanding of the technical topics like waveform conversion, sequencing, and noise generation. It will involve reviewing existing literature, tutorials, and research papers to gather relevant information and insights and to identify best practices and design principles for synthesizer circuit design.

2. Circuit design and implementation:

In this phase, the knowledge gained from the research will be applied to design the synthesizer circuits to implement on the LABOR workstation. This will involve creating circuit schematics, selecting appropriate components. The design process will be iterative, with refinement of the circuits based on observed performance and theoretical expectations.

3. Testing and evaluation:

This phase will involve testing the implemented circuits on the LABOR workstation, measuring their performance, and comparing the results with theoretical expectations. The testing process will include using oscilloscopes and other measurement tools to observe the generated signals. Each circuit section will first be examined independently so that possible problems can be identified more easily and depending on the circuit, data will be measured which include waveform shape, signal amplitude, timing accuracy, frequency response, and noise behaviour.

4. Software integration:

In this phase, software tools will be developed to facilitate the testing and analysis of the circuits which include programming and designing the circuit using specific software tools. The software will be used to visualize the generated signals, perform signal analysis.

5. Documentation:

The final phase will bring together the results from the previous work packages and involve documenting the entire design process, circuit schematics, images, experimental results, and analysis in a clear and educational manner. The thesis will describe the development process and discuss both the successful results and the limitations encountered during the whole process.

1.4. Work Plan

Use a Gantt-Chart with Mermaid, see: - [Quarto - Diagrams - Mermaid - Gantt](#)

Month 1: Technical research, literature review and analysing Moritz Klein's tutorials. Month 2: Familiarization with the LABOR workstation, initial circuit design, and circuit calculations. Month 3: Circuit implementation on LABOR and testing of the individual circuits. Month 4: Further testing, evaluation, and refinement of the circuits based on observed performance. Month 5: Development of software tools for testing and analysis, and integration with the hardware circuits. Month 6: Documentation of the entire design process, circuit schematics, experimental results, and analysis.

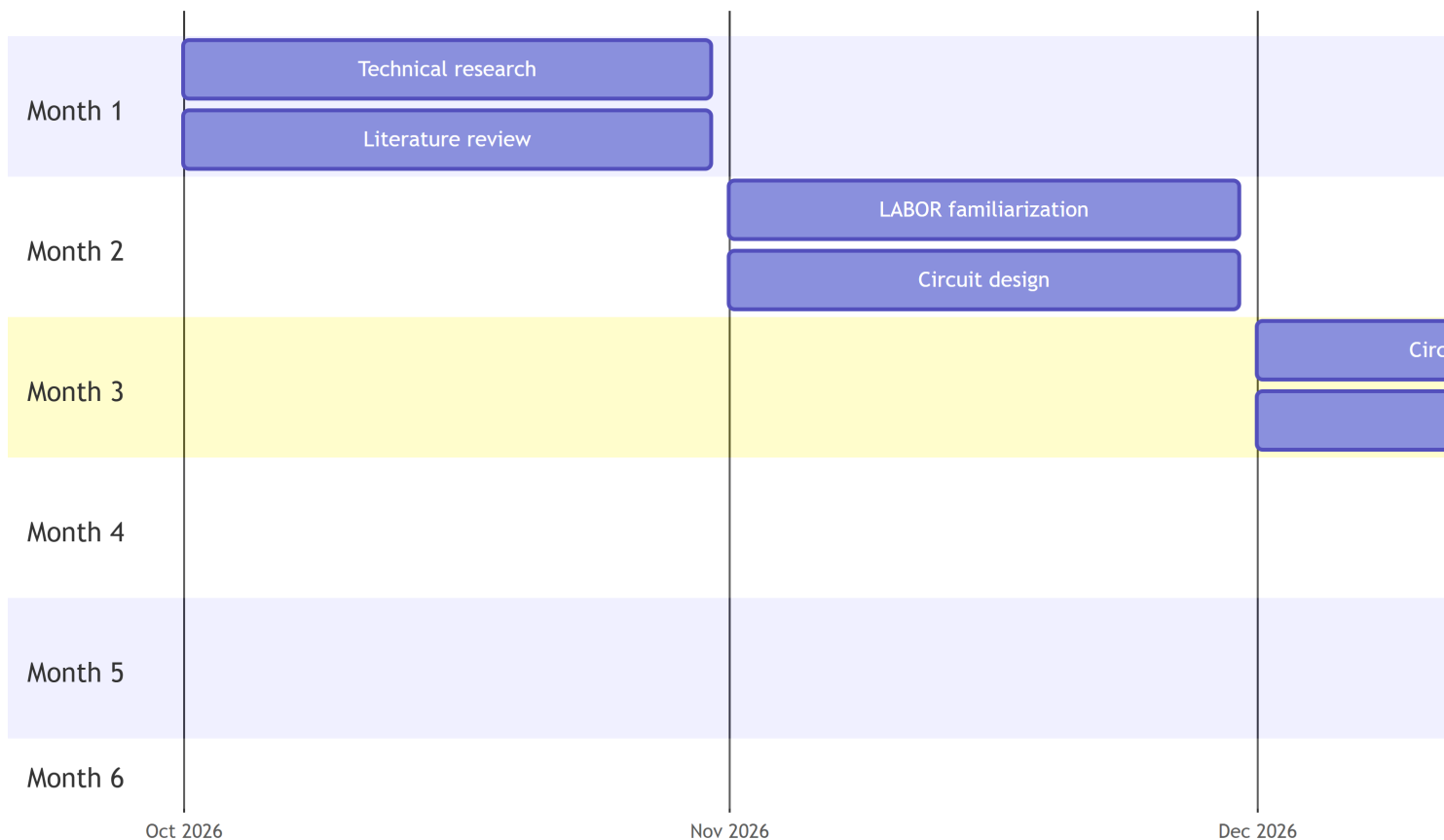


Figure 1.1.: Project timeline of the thesis work

1.5. References

Put them all in your bibliography (BibTeX file `__bibliography/references.bib`), use a reference manager like zotero and then cite. This will automatically generate a bibliography at the end of your thesis, see `references.qmd` file.

%%See: - [Quarto - Citations](#) - [Quarto and BibTeX](#) - [Zotero](#) and [Better BibTeX](#)

1. Modelling the Excess Noise due to Avalanche Multiplication in (Hetero-Junction) Bipolar Transistors - J. C. J. Paasschens and R. de Kort - Bipolar/BiCMOS Circuits and Technology, 2004.
2. Probability, Random Variables and Stochastic Processes - A. Papoulis, S. U. Pillai - 4th ed., McGraw-Hill, 2002.
3. The Differential Pair as a Triangle-Sine Wave Converter - R. G. Meyer, W. M. C. Sansen, S. Lui, S. Peeters - IEEE Journal of Solid-State Circuits, June 1976.
4. Design with Operational Amplifiers and Analog Integrated Circuits - S. Franco - 4th ed., McGraw-Hill, 2015
5. Limitations of a relaxation oscillator in capacitance measurements - Y. Liu, S. Chen, K. Nakayama, K. Watanabe - IEEE Transactions on Instrumentation and Measurement (Volume: 49, Issue: 5), October 2000.

6. Tunable Waveform Generator for Sensor and Instrumentation Applications - Sougata Kumar Kar, Siddhartha Sen - IEEE Transactions on Instrumentation and Measurement (Volume: 60, Issue: 10), October 2011.
7. <https://www.moritzkleinstruments.com/>

References are not only a list at the end of your report, they need to be linked/referenced from your writings.

Chapter 2.

Method

This report underscores the transformative impact of reproducible research on the scientific landscape, promoting a commitment to openness and accountability for the benefit of the entire research community.

References

- [1] M. Klein, “LABOR.” [Online]. Available: <https://www.moritzkleinstruments.com>
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- [6] J. C. J. Paasschens and R. de Kort, “Modelling the excess noise due to avalanche multiplication in (hetero-junction) bipolar transistors,” *Institute of Electrical; Electronics Engineers (IEEE)*, 2004, pp. 108–111. doi: [10.1109/BIPOL.2004.1365757](https://doi.org/10.1109/BIPOL.2004.1365757).

Appendix A.

Tools

This thesis was written using [Quarto](#)

Appendix B.

USB-Stick and Script Referencing

All directory structures and file paths presented in this appendix are intentionally simplified. They are provided to clarify the organization of the project and to indicate where specific scripts, netlists, verification outputs, and layout files can be found within the design environment.

```
thesis
  .gitattributes
  .gitignore
  _book/
  _quarto/
  _quarto.yml
  _bibliography
  index.qmd          # Main project page
  README.md
  references.qmd
  appendix           # Appendix
```